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The IT Identity Threat: A Conceptual Definition and Operational Measure

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ABSTRACT: As individuals' relationships with information technology (IT) grow more complex and personal, our understanding of the problem of resistance to IT continues to evolve. Current approaches to resistance are based on perceived threats to work tasks and social structure. This work enhances our understanding of resistance by developing a definition and measure of the IT Identity Threat, a new construct that integrates social, task-related, and personal factors of resistance. Grounded in identity theory, the IT Identity Threat offers a parsimonious means to explain and predict IT resistance behaviors. Using data from two independent studies conducted among students and faculty at a large university in the

Southeastern United States, we validate an operational measure of IT Identity Threat as a second-order construct and demonstrate that it successfully predicts resistance to IT. Our findings provide support for the IT Identity Threat construct as a simple mechanism to study resistance to IT.

KEY WORDS AND PHRASES: IT Identity, Identity Theory, Resistance to IT, Self-Esteem, Identity Threat.

Introduction

The faculty meeting has ended and Professor Patricia Nims is not happy. The department is adopting “online teaching,” and starting with the next fall semester, all faculty will be required to teach at least three credit hours per year through an online classroom management system instead of lecturing in person. “Professor Pat,” as her students call her, sees this as a major threat to the way she lives her life as a faculty member. Will online lectures give the university administration the capacity to record and distribute her most valuable work? And what will that do to her value as a professor, once a database of her lectures is available online? Will she still be able to achieve the level of excellence she is famous for, when she is forced to interact through an online tool? Moreover, Professor Pat feels that using online tools to communicate reflects a lack of effort. To her, taking the time to talk in person is an important sign of how much she values other people. Considering this, she is inclined to avoid and disparage the new system. However, she still loves her job and believes in her ability to overcome any challenge. Considering all these factors, what will she do?

The IT Identity Threat is a new construct that offers a theory-driven, parsimonious, and forward-thinking approach to resistance to information technology (IT) in the workplace. Resistance to IT is a persistent problem in industry, despite extensive research in the information systems (IS) field devoted to understanding its causes [2]. Organizations invest in IT to gain advantage [19]. However, any advantage gained from IT investment depends on individuals making use of a successfully implemented IT [1, 40]. Resistance to IT threatens implementation and use, and thus jeopardizes the value organizations might realize from IT investment. In the case of Professor Pat, her work experience makes her a particularly valuable member of her organization. However, her value may be jeopardized by how she reacts to the online teaching system, because she may not adapt and use the new system. In this case, the implementation of an IS may harm, rather than help, an organization.

To understand this issue, IS research has examined the IT artifact’s impact on work tasks [48], and has also looked at social factors that influence the acceptance of [41] or resistance to IT [47]. However, studying task-related and social factors in isolation is problematic, because a given IT’s effects on social structure and on

work tasks are difficult to parse from one another [54]. For example, it is generally recognized that resistance to IT is not fully explained by either poorly designed artifacts or by an IT's social effects alone [23]. Rather, perceptions of the IT artifact's impact on work must be integrated with its social consequences in order to explain the problem of resistance [52].

Where prior work has extensively examined how to conceptualize and measure why individuals accept IT, substantially fewer papers have attempted to create a means to study the social and task-related factors of resistance in concert, based on a parsimonious and measurable construct. For our approach, we draw from the rich identity literature and argue that the "IT Identity Threat" is a missing and critical integrative construct that can explain resistance behaviors. As implied by the name, this construct and measure is based on how ITs threaten individuals' identities: sets of beliefs about themselves as worthy members of a social group, as valuable and competent workers, and as unique and idiosyncratic human beings. These beliefs determine how individuals assess themselves as people, that is, determine their levels of self-esteem; thus, a threat to these beliefs may be a potent source of protective behaviors.

Self-esteem is a long-studied and reliably measurable construct that can be used to apply theory across a wide range of contexts [84]. The IT Identity Threat offers a parsimonious measure of the effects of a wide range of perceptions, because it captures the ultimate indicator of an individual's perceptions of an IT: namely, levels of self-esteem as a user of that IT. In other words, the basic premise of the IT Identity Threat is that individuals realize harm from an IT when they like themselves less due to using that IT, and the anticipation of further harm causes them to resist it. This is a broad premise, but the IT Identity Threat affords a model based on the integration of the broad range of self-esteem perceptions into a single construct. Thus, it answers the call for theory that is broad in scope, yet parsimonious in conceptualization [6].

To realize the IT Identity Threat's potential to explain individual reactions to a wide range of perceptions, this study addresses two research questions: "*What is the IT Identity Threat?*" and "*How can IT Identity Threats be measured?*" In examining these questions, we also *assess the predictive validity of the construct by empirically examining its relationship with resistance*. Ultimately, our goal is to use the IT Identity Threat to explain why people like "Professor Pat" may resist an IT, and set the stage for a new understanding that may help her adjust her identity rather than resist the IT.

We begin with an explanation of identity and follow that with a discussion of threats and resistance to IT. This leads to an explanation of the threat-based approach to resistance, identity threats, and how ITs can cause identity threats. Next, we define the IT Identity Threat. Then, we develop a second-order conceptualization and measure of the IT Identity Threat based on reduced self-esteem. To evaluate that measure's reliability and validity, as well as the second-order structure of the IT Identity Threat, we then present the results of two studies of individuals

who have used potentially threatening ITs in distinct contexts. Finally, we discuss our work's contributions, implications, and proposed directions for future research on the IT Identity Threat.

Defining Identity

"Identity" refers to the sum of beliefs held by an individual about who they are ("self-beliefs"). Identity embodies individuals' relationships with others, the roles they hold, and their personal historical narratives [17]. As such, identity is thought to cause a wide range of behaviors that individuals employ in the attempt to "enact" identity, that is, perform actions that harmonize with who they believe themselves to be [3]. Thus, our vignette's Professor Pat may enact behaviors that harmonize and reinforce her beliefs about: 1) how she relates to students and other faculty, 2) how she fulfills her role as a teacher, and 3) her own personal history, which has culminated in her uniqueness as a human being.

Individuals are thought to hold identities of three types: social identities, role identities, and person identities [17]. Social identity is the sum of beliefs about the self as the member of a group (an "in-group," which may be contrasted with "out-groups," or groups the self does not belong to) [38]. Membership in an in-group that is associated with high levels of power, prestige, and resources causes positive feelings about the self [91]. Specifically, it raises an individual's "self-worth," reflecting the value they offer as a member of an in-group [17, 21]. Experiences that confirm beliefs in the power and prestige of an in-group reinforce beliefs that an in-group is powerful and prestigious. Experiences that confirm membership in an in-group strengthen beliefs that an individual embodies the power and prestige of that in-group. As these beliefs grow in strength, perceptions of the self as worthy and valued grow. Conversely, experiences that reduce individuals' beliefs in the power and prestige of their in-groups, or the extent that they themselves embody the power and prestige of their in-groups, should lower this form of self-esteem. Thus, levels of social identity are thought to be reflected by levels of worth-based self-esteem [84].

Role identity refers to the part of the self that is composed of a role the individual plays in social contexts [86]. For example, an individual who teaches has a teacher role identity. Role identity is not just the belief that an individual performs a role; it also encompasses beliefs about how well the individual performs the tasks associated with that role (i.e., competence) [13]. This is because the successful enactment of tasks in the domain of a role identity serves as evidence of the validity of that role identity [17, 82]. It may help to imagine an individual who holds a computer programmer identity. Successfully completing coding projects would confirm self-beliefs in the programmer role identity, and these self-beliefs would contribute to perceptions of competence in the domain of computer programming. Because of this, role identity is reflected by levels of self-esteem based on competence [21].

Finally, person identities consist of beliefs about the self that distinguish the self from others [81]. Beliefs regarding the self as a unique and idiosyncratic individual make up one's person identity, and the sum of beliefs and characteristics that distinguish the self as an individual comprise the idealized self [17]. Individuals seek experiences that give them a sense of continuity regarding their person identities, and they try to behave in ways that harmonize with an "authentic self" that is consistent across contexts [88]. Because of this, person identity is associated with self-esteem based on how true an individual is to his or her authentic self [84].

A single individual holds many identities; for example, Professor Pat may view herself as a member of a university faculty (social identity), as a course instructor and advisor to students (role identities), and as a unique and authentic individual marked by a friendly demeanor (person identity) [17]. The three types of identity are summarized in Table 1.

Over time, identities gradually change in a cyclical manner, in response to the life experiences of the individual. Experiences and behaviors that are consistent with an individual's self-beliefs strengthen those self-beliefs, confirming identity and contributing to self-esteem [83]. This increase in self-esteem, in turn, causes individuals to seek out experiences and behaviors that are consistent with their identities [17]. This is a cycle of self-esteem and identity building; it is valuable, and individuals work to maintain it [88]. For example, when a student enthusiastically hails the professor in our vignette as "Professor Pat," some of Professor Pat's self-beliefs are confirmed, and thus her identities are verified and strengthened. First, there is the recognition of her as a professor, a member of the faculty (social identity), and an instructor (role identity). Second, the student's selection of "Professor Pat," instead of the more formal "Professor Nims," is evidence of the friendliness that marks Professor Pat as a unique and authentic individual (person identity). As a result of this experience, Professor Pat benefits from elevated levels of self-esteem, and will thus select behaviors that are likely to result in similar experiences.

By contrast, individuals avoid experiences that conflict with their identities [68] in order to maintain levels of self-esteem. To extend our vignette's example, the idea of online teaching may harm Professor Pat's social identity as a faculty

Table 1. Types of Identity

Identity Type	Theoretical Basis	Example
Social: composed of beliefs about the self as the member of a group.	Haslan 2004 [38]	Pat is a valued member of the faculty.
Role: composed of beliefs about the self as fulfilling a social or work-related role.	Burke 1991[13]; Stryker and Burke 2000 [86]	Pat is a great teacher who engages well with her students.
Person: composed of beliefs regarding characteristics that distinguish the individual from others.	Stets and Burke 1994 [81]; Burke and Stets 2009 [17]	Pat is the friendly and authentic "Professor Pat."

member, because it involves a database of lectures that could be used without the presence of an actual professor. Such a database threatens the value of faculty, and thus reduces Pat's perceived value as a member of the faculty. Moreover, she may doubt how well she will embody her instructor role identity through an online medium: this would reduce the self-esteem she feels as a competent instructor. Finally, Pat's beliefs about herself as a unique and authentic individual may depend on the emotional warmth of in-person communication, as opposed to online interaction. Thus, using an online teaching system may seem inconsistent with her authentic self. As a result of these concerns, she may perceive the online teaching system as a threat.

Threats and Resistance to IT

Research on resistance to IT often examines why people may view an IT as a threat, either to work outcomes or to a social group's status or power. For example, Marakas and Hornik [57] proposed that fear associated with the inability to accomplish work with a new IT causes individuals to resist that IT. In that study, individuals avoided the IT and sought out familiar (although obsolete) work routines. Similarly, Kim and Kankanhalli [44] proposed switching costs to a new IT as a threat to a desirable status quo, finding that individuals resist an IT based on the ratio of the benefits to work outcomes vis-à-vis the time and effort required to adapt to the IT. They also hypothesized and found support that colleague opinion of an IT affects levels of resistance to that IT, recognizing individuals' desire to conform to their social group. Lapointe and Rivard [47] offered a perspective that combines social and work-related factors. They found that individuals may initially respond to perceptions that an IT causes negative work outcomes, and when enough individuals share these perceptions, a social factor of resistance emerges.

Some suggest that resistance to IT may result from a two-stage appraisal process, in which a primary appraisal is made to determine whether an IT presents a threat or opportunity, and then a secondary appraisal determines whether the situation may be controlled so that the threat can be averted or the opportunity exploited [8]. From this perspective, the anticipation of harm to work outcomes or social status causes an IT to be perceived as a threat to be resisted, while managerial support for an IT makes that IT seem helpful for career growth, and thus a source of opportunity to be exploited. The secondary appraisal concerns the individual's ability to control their situation regarding the IT. Generally, it is thought that a high level of control is likely to increase perceptions that an IT may be beneficial and exploited. From a resistance perspective, however, control over the situation also makes it possible to avoid using the IT, or to use the IT while resisting change to work routines, thus negating much of the value the organization likely expected from investment in that IT [11,48].

The IT Identity Threat complements research that focuses on assessments of an IT as a threat, and advances it in two ways. First, it explains why individuals may

resist an IT for completely personal reasons, rather than focusing only on the effects of an IT on social structure and work tasks. ITs are becoming more intrusive and personal in their use [20]. In an extreme manifestation of this, some firms now implant computer chips inside their employees [36]. It is easy to imagine that such an implanted IT may help with some work tasks, but still attract intense resistance. As technology continues to change how workers view themselves as people, it is critical that our field develops an understanding of why a technology may be resisted for completely personal reasons.

Second, the IT Identity Threat expands our current understanding by offering new avenues for treatment. Existing work suggests responding to user feedback to alter perceptions of the IT as a threat (or opportunity) and encouraging use by altering perceptions of managerial support for the IT [44, 47]. However, these treatments may not always be feasible. In some cases, threatening perceptions of an IT become established before management can intervene. Additionally, since software designers must often surrender ease of use in order to implement important features [29], some systems may, by necessity, be challenging for new users because of the breadth and depth of the IT's functionality. Finally, existing works (e.g., that of Bala and Venkatesh [7]) point to managerial support as a treatment for resistance. However, negative perceptions of a system may be the *result* of perceived managerial support for the IT [31, 54]. This happens when individuals see the IT as a tool that increases management's control and power at their expense [58]. In this kind of situation, increasing the prominence of managerial support for the IT may exacerbate resistance. The IT Identity Threat expands our current state of research with identity-based treatments that may be effective even when no managerial intervention is likely to reduce perceptions of threat (as may often be the case). This is because it is based on the identity threat framework, which is not only explores how to reduce perceptions of threat, but also focuses on finding treatments for threats that are inevitable [68].

Identity Threats

In the workplace, many experiences can challenge the self-beliefs that comprise identity. When an experience contradicts identity, individuals experience a loss of self-esteem and take action to preserve the self-esteem associated with identity. This is the nature of the identity threat, defined by Petriglieri as "experiences appraised as indicating potential harm to the value, meanings, or enactment of an identity" [68, 644].

The pervasive nature of IT in the workplace makes it a potent source of experiences that may comprise identity threats. ITs impact work tasks and social structure, and can cause individuals to change the way they work and achieve goals. As such, IT's change what individuals do. Changing what an individual does may generate experiences that challenge, rather than confirm, the self-beliefs that comprise identity. For example, if a hospital adopts an IT, and that IT changes how nurses

interact with patients, then that IT has the capacity to change the social status of nurses, the work of nurses, and even what it *means* to be a nurse in that hospital [63]. In this way, an IT may change *who* the individual is, that is, their *identity* as a nurse. The new, IT-infused, meaning of “nurse” may not appeal to some nurses. Thus, an IT may force unwanted changes to the “nurse identity,” resulting in an IT Identity Threat that would cause resistance to the IT [93].

How individuals react to such threats is described by the identity threat framework [68]. The identity threat framework provides an abstract process model that explains individual reactions to a broad range of stimuli that conflict with the self-beliefs that comprise identity. The basic premise of the identity threat framework is that individuals experience stress from these stimuli, anticipate further stress, and are thus forced to cope with the source of such stimuli as a threat. As is the case in much of the threat literature, this framework predicts that individuals will either resist the source of identity threat or adapt to defuse the source of identity threat [63]. The identity threat framework has served as a basis for research on a wide range of topics, such as organizational identification [5] and recovering from professional setbacks [80]. Information systems research has leveraged the identity threat concept to theorize about individual reactions to IT. Specifically, it has been proposed that exposure to an IT may cause a discrepancy between self-beliefs and workplace reality, resulting in negative emotional arousal that causes a wide range of behaviors, from resistance to the undertaking of personal change [64]. IS research has also leveraged the concept of identity threats to explore the challenges offshoring presents to software developers [45].

IT as a Source of Identity Threat

Per the identity threat framework [68], intergroup conflict, obstructions to identity verification,¹ or changes to identity meaning may harm identity. This harm causes individuals to experience stress and anticipate further stress based on potential future harm to identity. To cope with that stress, they take action to diffuse the source of threat (Figure 1).

Intergroup conflict may cause a group to lose resources and status, and thus harm self-beliefs regarding the power and prestige an individual realizes from their social identity as a member of that group. Changes to work tasks that had previously facilitated the enactment of identity may threaten self-beliefs regarding how much an individual embodies their role identity. Finally, if the tasks assigned as part of a job change, then the meaning of that job is also changed. In such a case, the new meaning of the job may not be consistent with person identity. The result of any of these experiences would be a threat that causes individuals to respond, ideally by disarming the source of the threat through resistance [68]. This process is further contextualized to the domain of IT in the following paragraphs.

In the workplace, intergroup conflict may result from an IT changing the distribution of power in the organization. Inter-group conflict, one of the hallmark

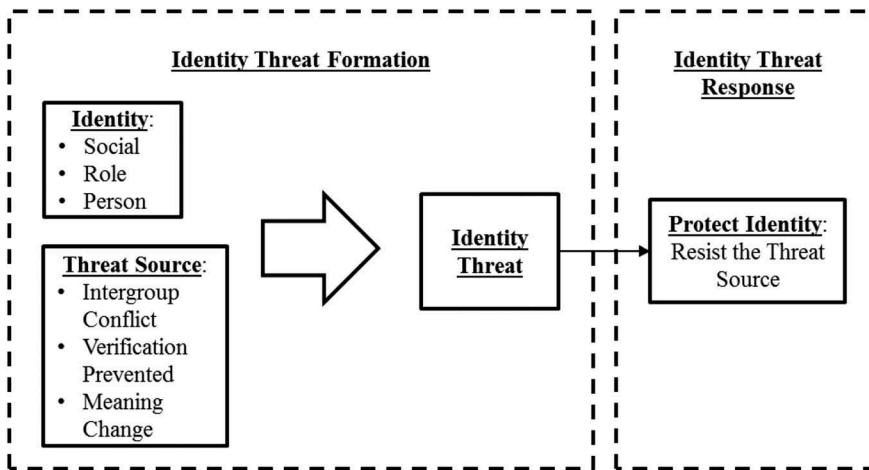


Figure 1. Process Summarized from the Identity Threat Framework (based on [68])

sources of identity threat [69], is associated with IT implementations, often resulting in the loss of power and prestige associated with membership in a social group [54]. Social identities are composed of beliefs about the self as an individual member of a social group [17]. Thus, when an IT reduces the power and prestige associated with membership in a group, it may threaten the social identity held by an individual in that group. For example, in one study of physicians who were asked to use managerial software for a significant portion of their work, physicians were asked to adjust their job description from clinical physician to “doctor as manager.” Given the dramatic image of the medical community as compared to the more prosaic image of managers, some physicians may have perceived their membership in the “physician” social group as less prestigious and resisted that change [31]. Likewise, consider Markus’ classic study of an IT’s impact on the relative power of accountants at different levels in a large corporation, which describes resistance as the group-level result of the loss of power and authority by a group [58]. An identity perspective may help conceptualize the same phenomenon on an individual level. Membership in a powerful or prestigious group generates favorable feelings about the self [91]. Conversely, any phenomenon that reduces the power of a group may cause a loss of favorable feelings about the self as a member of that group. When an IT causes a group to lose power and prestige, individuals who identify as parts of that group may experience harm to their own individual social identities. Thus, the individual accountants observed by Markus [58] may have resisted the technology because it presented a threat to each of their individual identities as members of a group.

Likewise, the prevention of identity verification in the workplace may be the result of an IT’s impact on work tasks. The competent execution of work tasks provides evidence that an individual embodies the role identity associated with those tasks. Because of this, the competent execution of tasks associated with

a work role is critical for identity maintenance. An IT can threaten the competent execution of tasks while individuals adjust to IT-enabled work routines, possibly due to the learning curve associated with new work tasks [44]. In the process of overcoming this learning curve, individuals will experience less validation of their role identities. In this way, the IT prevents the verification of role identities; this is a hallmark of identity threat sources [68]. To verify and thus restore beliefs in competence, individuals may seek to avoid an IT and its associated work routines, seeking instead to enact other, possibly obsolete, work routines [57]. For example, in a 2004 study of physicians by Doolin [31], physicians viewed an IT implementation as a tool of management to increase efficiency at the possible expense of clinical effectiveness, which is critical to the physician role identity. As a result, many of the physicians resisted, refusing to enter complete data into the system and continuing to use their own individual PC-based systems. Similarly, Marakas and Hornik [57] described a group of managers confronted with new IT-enabled work routines. These managers experienced stress associated with the new work routines. They coped with that stress by avoiding the new work routines and engaging in obsolete, but familiar, work routines [57]. From an identity perspective, these managers' behaviors can be interpreted as attempts to resume the role identity verification associated with those familiar work routines.

Finally, an IT may threaten person identity. IT use has been shown to change person identity. For example, the use of smart phones may influence self-beliefs about conscientiousness, because smart phones increase responsiveness, as well as expectations of responsiveness, in communication [20]. As such, ITs can change the idiosyncratic nature of the individual, that is, change their person identity. Such a change might not be beneficial, especially if it is not consistent with the idealized self. It could threaten the continuity that supports the belief in an authentic self and be resisted, even if it helps the individual overcome an admitted deficiency [88]. For example, an individual who believes that he or she is forgetful may resist the use of a memory aid such as Google Glass (computer-enhanced spectacles) because wearing a computer on the face may be too inconsistent with their authentic self. Rather than become "the type of person who would wear Google Glass," this individual resists Google Glass and maintains authentic selfhood.

Similarly, the use of an IT in the workplace may eliminate some of the most rewarding aspects of work through automation [89] and thus change the meanings associated with an identity. For example, take a hypothetical case of a workshop enforcing a rule that woodworkers must use computer-controlled lathes instead of working by hand. In such a case, using the computer-controlled lathes may change the meaning of being a woodworker from being one who works with wood, to one who works with computers.

The identity threat framework proposes that individuals will react to an identity threat primarily by engaging in protective behavior. Individuals may attempt to defuse an identity threat by acting or speaking in opposition to the

threat source [68], and by praising the threatened identity [33]. In the hypothetical woodworking example, protective behavior may take the form of deriding the quality of the computer-controlled lathe's output, and of praising the quality of hand-worked goods.

However, resistance to IT is not the only possible outcome of the IT Identity Threat. An identity threat may be defused by changing the self rather than trying to avoid or resist a threat source [68]. Individuals may even adopt a new identity that incorporates the technology into their work lives [64]. However, the extent to which individuals will try to change the self rather than resist a threat source depends on the extent to which they believe they are capable of undergoing such a change, that is, their levels of self-efficacy for change [68].

Defining the IT Identity Threat

The IT Identity Threat captures what individuals perceive as social, work-related, and personal threats that cause them to resist an IT. To define this construct, we first consider its conceptual domain and demark the entity it applies to [55]. Conceptually, the IT Identity Threat is a threat, distinct from other threats associated with IT because it is a threat to identity. Prior research has conceptualized *threat* as the anticipation of harm [62]. *Identity* has been conceptually defined as the sum of an individual's self-beliefs [17]. Thus, the IT Identity Threat is defined as *the anticipation of harm to an individual's self-beliefs, caused by the use of an IT*, and the entity it applies to is the individual user of an IT. IT Identity Threats are evoked when individuals who use an IT anticipate harm to their identities, reflected by lower levels of self-esteem in their assessment of identity.

Assessment of the IT Identity Threat

The latent nature of identity makes it difficult to measure. The prevailing view in the identity literature is that individuals perceive changes to identity as changes in self-esteem, and are not directly aware of changes to identity [17,21,53]. Identity is a collection of self-beliefs, while self-esteem is the individual's judgment of the self, based on those self-beliefs [84]. The self-beliefs that comprise identities are independent attributes of the self, and each self-belief contributes to the individual's overall self-esteem in its own way [27, 53]. For example, Professor Pat's self-beliefs of being a valued faculty member, a good teacher, and a friendly person all contribute to her self-esteem. Experiences that counter the self-beliefs that comprise identity reduce self-esteem. Therefore, reduced self-esteem is an indicator of harm to identity. Thus, measurement of the IT Identity Threat is based on perceptions of reduced self-esteem caused by an IT.

Dimensions of the IT Identity Threat, Reflected by Change in Self-Esteem

The three types of identity are often treated as independent, but related, constructs [17], each reflected by its own type of self-esteem (worth-based, competence-based, and authenticity-based) [84]. Because of the breadth of changes that ITs are known to cause [54], ITs have the power to impact and, thus, threaten, all types of identity: social, role, and person. Accordingly, the full conceptualization and measurement of the IT Identity Threat must consider the IT's potential to harm identities of all three types. However, the IT Identity Threat's dimensions do not reflect a latent "threat-ness" that necessarily causes the loss of all three types of self-esteem. It is possible for an individual to experience threat when faced with reduced self-esteem of a single type. Even though a single identity may be related to identities of other types of identity [17], some experiences are thought to have effects that vary across types of identities, or only effect one type of identity. This has been observed in studies of such identity-challenging events as marriage and childbirth [15] where role identity (e.g., wife) may be affected more than other identities. Because experiences may not equally affect all identities, they could differentially affect one dimension of the IT identity threat over another (e.g., competence-based self-esteem could be affected more than authenticity-based self-esteem). Therefore, we argue that the three dimensions of the IT Identity Threat will not necessarily co-vary across contexts. This is a defining characteristic of an aggregate second-order construct [73]. Conversely, the indicators of each type of self-esteem are thought to be reflective [39]. Accordingly, we formally specify the IT Identity Threat as a second-order formative (aggregate) construct, with reflective first-order dimensions [73].

The first dimension of the IT Identity Threat captures anticipated harm to social identity. While social identity is the result of identification with an in-group [38], changes to the status of that in-group are experienced as changes to self-worth on an individual level [91]. This is because perceptions belonging to a powerful and prestigious in-group are reflected by how worthy and valued individuals believe themselves to be in the eyes of others [21]. Based on this, we propose that at least part of the IT Identity Threat, as perceived by individuals, would be a loss of self-worth. Thus, we define **loss of worth-based self-esteem** as *a reduction of how worthy and respected the individual feels they are as a member of a social group* [53]. Experiences that threaten social identity would thus harm beliefs about the self as a member of a social group, and beliefs about the self as a member of a social group are thought to manifest as worth-based self-esteem [78], based on the value that the individual offers to others [21]. Thus, this dimension concerns a reduction in the part of an individual's worth-based self-esteem that depends on beliefs about membership in a social group [53].

The second dimension of the IT Identity Threat captures harm to role identity, and is based on how competently individuals believe they perform the tasks associated with their role identities. Under nonthreat circumstances,

work routines reinforce identity through the rewards and reflected appraisals that come with the successful accomplishment of work goals. Individuals then invest themselves in those work routines in a cycle of work, achievement, and identity reinforcement [16]. This cycle contributes to role identity, which is manifested by competence-based self-esteem, i.e., the belief that one executes their role-based tasks well [21]. An IT that interrupts this cycle of role enactment threatens the role identity and threatens the beliefs of competence tied to role identity. Thus, anticipated harm to role identity would be manifested as reduced competence-based self-esteem. Based on this, we define **loss of competence-based self-esteem** as *a reduction of an individual's belief in their ability to perform work* [26].

Finally, we propose a dimension of the IT Identity Threat that captures anticipated harm to person identity. The desire to enact behaviors that are consistent with an individual's authenticity, based on "one's internal or personal standards as to who one really is" [84, p. 411] is a powerful source of motivation [88]. When an individual uses an IT that is not compatible with person identity, that individual is behaving in ways inconsistent with his or her authentic self [16]. Authenticity-based self-esteem reflects the extent to which individuals see themselves as unique and idealized individuals [14] and would, therefore, be reduced when an individual believes that he or she has become less like the idealized self. Thus, we define **loss of authenticity-based self-esteem** as *a reduction of the individual's belief in the possession of traits and personal characteristics associated with the idealized self* [16].

Each the three dimensions of the IT Identity Threat reflects the perceptions that individuals experience when exposed to one of the three threatening phenomena identified by the identity threat framework [68]. Intergroup conflict reduces the power and prestige of the in-group, resulting in a loss of the worth-based self-esteem felt by individuals in the in-group. Prevention of identity enactment reduces the beneficial outcomes normally experienced from familiar work routines, and this causes a loss of competence-based self-esteem. Finally, a change in the meanings that underlie a workplace identity may interrupt the coherence of an authentic self; this would be perceived as a loss of authenticity-based self-esteem. The manifestation of these threat sources is summarized in Table 2.

Table 2. Manifestation of IT Identity Threat

Threat Source	Harm Manifestation	Perception of Harm
Intergroup Conflict	Loss of In-group Power and Prestige	Loss of Worth-Based Self-Esteem
Prevention of Enactment	Reduced Beneficial Work Outcomes	Loss of Competence-Based Self-Esteem
Meaning Change	Lack of Coherence	Loss of Authenticity-Based Self-Esteem

Development and Validation of the IT Identity Threat Measure

To develop and validate our measure, we followed guidance from decades of construct development research in the IS field [25,55,85]. After conducting a literature search to generate an initial set of items, we carried out a series of exercises designed to refine the measure. This process concluded with a pilot study and rigorous tests to evaluate the nomological validity of the measure, along with tests of its measurement model.

Item Generation and Refinement

Our first step was to find items from previous studies of self-esteem and identity and adapt them to fit an IT context [16, 20, 30, 34, 39, 53, 77]. To ensure that our items represent the meaning of our dimensions, we subjected our pool of initial items to a series of four card-sorting exercises [55]. During each round of these exercises, a diverse group of judges was asked to assign each measurement item to one (or none) of our three dimensions [61]. Each judge was given the names and definitions of our constructs and worked alone, with each of the items written on individual cards, presented in random order. After each exercise, items that failed to achieve a moderate level of agreement were either reworded or eliminated [46]. No judge participated in more than one exercise, and the final sorting exercise was carried out by 22 judges, the majority of whom were working professionals. The demographics and professions of these judges were: 8 female, 14 male, aged from 26 to 59; 9 PhD candidates, 1 professor, 4 engineers, 2 small business owners (1 in retail computing and 1 in publishing), 2 computer programmers, 1 lawyer, 1 auditor, and 1 executive director of development at a Fortune 50 company. At the conclusion of this phase of analysis, seven items were retained for each dimension (found in Appendix A).

Pretest and Pilot Test

Following the card-sorting exercise, the remaining items were subjected to a pretest to evaluate the reliability of our items and a pilot test to evaluate the conceptualization of the IT Identity Threat [85]. Our population of interest consists of individuals who 1) would be required to use an information system and 2) may anticipate harm from that system. We therefore selected a sample frame of undergraduate students who 1) were forced to use an online portfolio system in order to graduate and 2) who might perceive social, role-based, and personal impacts from the use of that system. For our pretest, we collected data from 200 undergraduate students from a university located in the US. Qualitative feedback, as well as tests of reliability resulted in a shorter instrument that was then used in a pilot test (see Table 3).

Table 3. Measurement Items for the IT Identity Threat

Construct	Measurement Item	Source
Loss of Worth-Based Self-Esteem: ePortfolio...	...makes me feel less respected by others in my peer group.	Fleming and Courtney 1984 [34];
	...people in my peer group will lose respect for me.	Rosenberg 1965 [77]; Burke and Reitzes 1991 [16];
	...others consider will consider me to be a poor member of my peer group.	Luhtanen and Crocker 1992 [53]
	...makes me feel that people in my peer group will admire me less.	
Loss of Competence-Based Self-Esteem: ePortfolio...	...makes me feel less confident that I understand things well enough to get work done.	Rosenberg 1965 [77]; Heatherington and Polivy 1991 [39]; Burke and Reitzes 1991 [16];
	...makes me feel less confident about having the skills needed to get work done.	Davis 2013 [30]
	...makes me feel that I have less ability to get work done.	
	...makes me feel that I do things poorly.	
Loss of Authenticity-Based Self-Esteem: ePortfolio...	...makes me feel displeased with who I am.	Rosenberg 1965 [77]; Burke and Reitzes 1991 [16];
	...makes me feel less like the person I want to be.	Heatherington and Polivy 1991 [39];
	...makes me feel discouraged with who I am.	Fleming and Courtney 1984 [34];
	...makes me take a less positive attitude toward myself as a person.	Carter and Grover 2015 [20]

Our pilot test ($n = 650$; sample frame: undergraduate students required to use an online portfolio in order to graduate from a university in the United States of America) provided evidence for a conceptualization of the IT Identity Threat as a second-order construct with three dimensions. The internal consistency of each dimension was supported by Cronbach's alpha, with all construct scores exceeding the recommended 0.7 [65]. Regarding factor structure, a test of competing structural equation models [18] using the software package EQS found support for a second-order construct with three dimensions. Using a X^2/df test of significance, a three uncorrelated factor structure showed a significant ($p < 0.05$) reduction of misfit over a single-factor structure. A second-order factor with three dimensions showed a significant ($p < 0.05$) reduction of misfit over the uncorrelated factor structure. Based on these results, we moved forward with formal studies to validate the IT Identity Threat as a predictor of resistance to IT.

Survey Analysis

With our pilot-tested instrument, we next conducted formal studies to evaluate our measure. Two cross-sectional surveys, consisting of Likert scale measures and administered via a Web site (Qualtrics.com) were used to gather data in two separate contexts. Items were identical across the two surveys, except for the names of the technologies involved. Results across disparate samples contribute to generalizability [43, 72]; thus, we selected two sample frames designed to improve the generalizability of our study to our population of interest. Our broad population of interest includes both students and working professionals, so we selected the contexts of undergraduate students and of faculty in a large public university² located in the US. As was the case in our pre- and pilot tests, the sample frame for one study (“Study 1”) was composed of undergraduate students who were required to complete work using a system called ePortfolio in order to receive their degrees. The sample frame for the other study (“Study 2”) consisted of university faculty who were required to use a system called iRoar to manage their class rosters, grade submissions, and other pedagogy related activities.

We selected these sample frames because each was likely to report different sources of identity threat from their respective mandatory ITs. In the case of students using ePortfolio, subjects during our pretest and pilot tests expressed concerns that: 1) the IT furthered the interests of certain administrators (the “ePortfolio people”) at the expense of students’ interests and was thus a potential source of intergroup conflict; 2) ePortfolio forced students to use an online publishing system to present their work, and many students found that system to be an impediment to the enactment of role-based tasks such as making progress toward graduation and seeking post-graduate employment; and 3) the ePortfolio system forced students to perform Web development, an activity that may have been inconsistent with their authentic selves. Students were so incensed by the system that they organized against it, posting an online petition for its removal that received thousands of signatures. Approximately one year after the completion of our data gathering, the university abandoned the ePortfolio system.

Similarly, the iRoar system was perceived by many professors to be a thoughtless intrusion on the behalf of the university’s administration, limiting their flexibility in how they accomplished work. In this way, it was the source of intergroup conflict. It also changed critical work tasks carried out by professors and, thus, may have prevented the enactment of some aspects of the professor role identity and may have had the potential to undermine the meaning of “professor” at the university by changing the work of the professor.

Participants were recruited via an email broadcast sent through university list services, and were motivated by the chance to win one of four tablets (approximate value of 450 USD each). After deleting unusable responses (e.g., those with no items answered) and outliers (see Appendix B for details of this process), we were left with 2,109 complete and usable surveys (out of an initial total of 2,759) for Study 1. This represents a response rate of 12.5% of the entire undergraduate

population of 16,391 during that semester. For Study 2, we received 203 complete and usable surveys. This represents a response rate of 17.5% of the complete population of 1,157 faculty during that semester.³ The demographic data for our subject pools are found in Appendix C.

Evaluation of the Measurement Model

For the sake of establishing nomological validity, we also tested the IT Identity Threat measure for correlations with similar constructs and as a predictor of resistance, along with a factor drawn from the identity threat literature [55]. Finally, we tested the dimensionality of our specified measurement model, that of a second-order formative construct with a reflective first-order [73].

Tests of Reliability

Because our items are reflective, they should all correlate highly within each of our dimensions [73]. Accordingly, we tested each of the IT Identity Threat's three dimensions for internal consistency. Cronbach's Alpha and Rho [76] were obtained from confirmatory factor analysis using the structural equation modeling software EQS [18]; each are presented in Table 4. All are higher than the cutoff of 0.7 recommended by Nunnally [65], indicating that the items for each dimension reflect common latent factors. The relative strength of correlation between each item and its construct is indicated by factor loadings, which are found in Appendix D.

Discriminant Validity

The purpose of these tests is to demonstrate the independence of our constructs from each other. We compared subjects' responses to the IT Identity Threat's items with those of other measures to demonstrate that the IT Identity Threat is a distinct construct and not a reiteration or mere inversion of the other constructs in its nomological net [55].

Discriminant validity is tested by comparing the correlations between constructs with the square root of the average variance explained of each construct obtained

Table 4. Reliability

Dimension	Study 1		Study 2	
	Alpha	Rho	Alpha	Rho
Loss of Worth-Based Self-Esteem	0.97	0.97	0.964	0.96
Loss of Competence-Based Self-Esteem	0.96	0.97	0.943	0.94
Loss of Authenticity-Based Self-Esteem	0.98	0.99	0.977	0.97

through factor analysis [35, 55]. As seen in Appendix D among the causal factors of resistance (the IT Identity Threat, Lack of Value, and Colleague Opinion of the IT), no construct correlation is higher than the square root of any construct's AVE.

Test of Predictive Validity

The IT Identity Threat should predict resistance to IT, a construct that is often broadly and inconsistently defined [47]. Most works conceptualize resistance to IT as a form of individual opposition in the form of passively avoiding the IT and/or actively campaigning against its implementation [22, 44]. Because the identity threat literature proposes active and passive resistance to a threat source [68], this research applies a broad conceptualization of resistance to IT: *opposing an IS implementation by means of signals, actions, and inaction*. This construct is based on the works of Kim and Kankanhalli and also Keen on resistance to IT [42, 44].

Controlling for age and gender, the IT Identity Threat model explained 18.7% of the level of resistance to IT in Study 1, and 26.3% of the same in Study 2. As illustrated in Table 5, both of our studies provide evidence of predictive validity, with the level of resistance to IT increasing with the level of the IT Identity Threat.

Test of Nomological Validity

The IT Identity Threat measure was developed to extend research on resistance to IT, based on identity threats. Accordingly, it is expected to correlate with known factors of resistance, and also relate to factors associated with identity threats [55]. For this exercise, we selected two constructs commonly identified as antecedents of resistance (one positive, one negative) [44], along with a factor from the identity literature that is thought to reduce levels of threat and resistance behaviors [68]. First, we used the lack of value associated with using the IT, defined as *a negative overall evaluation of the IT based on a comparison between the costs and beneficial*

Table 5. IT Identity Threat Predicts Resistance to IT (Control Variables: Age and Gender)

Details:	B	Std. Error	Beta	T	Sig.
Study 1: Resistance to IT (Mean: 5.24, SD: 1.51), R ² = 0.187, n = 2109					
(Constant)	4.089	0.135		30.283	p < .001
The IT Identity Threat (Mean: 3.63, SD: 1.70)	0.372	0.017	0.418	21.248	p < .001
Study 2: Resistance to IT (Mean: 2.48, SD: 1.37), R ² = 0.263, n = 203					
(Constant)	1.172	0.407		2.879	p < .010
The IT Identity Threat (Mean: 2.38, SD: 1.17)	0.602	0.074	0.5	8.083	p < .001

changes associated with the outcomes of use [44] as a causal factor of resistance. Second, we tested colleague opinion of the IT, defined as *the individual's belief that their colleagues support the use of the IT*, for a negative correlation with the IT Identity Threat. Finally, we examined self-efficacy for change, defined as *the individual's belief that they can effect beneficial changes to the self*. This construct was selected based on the identity threat framework's proposition that individuals are more likely to adapt the self, rather than resist a threat source, when they believe that they are capable of undergoing change. Estimated with SPSS v.24, the IT Identity Threat correlated with lack of value, and negative correlations with colleague opinion of the IT and self-efficacy for change. All significance factors ("p values") were under 0.001. Our construct correlations are found in Appendix D.

Tests of Dimensional Structure

Identity theory proposes that identities may be interrelated across type. That is, an individual's social, role, and person identities may each be influenced by the others. As a construct with dimensions based on these three interrelated types of identities, the IT Identity Threat may be conceptualized as a single phenomenon composed of three interrelated dimensions. To evaluate the discriminant validity of each of these dimensions, as well as between each and the construct of resistance to IT, correlations among them were measured against average variances explained (Table 6). The average variance explained of each dimension exceeded that dimension's correlations; this is evidence that these dimensions are discreet from one another, and from the construct they are intended to predict.

To evaluate the relationships among the IT Identity Threat's three dimensions, we used structural equation modeling to test competing models. A single-dimension model, three uncorrelated dimensions, and three correlated dimensions were all measured for misfit using EQS 6.1.

Table 6. Dimensional Correlations and Average Variance Explained

Construct	1	2	3	4
Study 1				
1: Resistance	0.884			
2: Loss of Worth-Based Self-Esteem	0.315	0.970		
3: Loss of Competence-Based Self-Esteem	0.350	0.446	0.968	
4: Loss of Authenticity-Based Self-Esteem	0.411	0.461	0.779	0.986
Study 2				
1: Resistance	0.882			
2: Loss of Worth-Based Self-Esteem	0.361	0.97		
3: Loss of Competence-Based Self-Esteem	0.468	0.305	0.968	
4: Loss of Authenticity-Based Self-Esteem	0.393	0.432	0.520	0.986

The bold text indicates the Average Variance Explained.

We tested for X^2 difference to evaluate the conceptualization of the IT Identity Threat as a single multidimensional construct, as opposed to a single one-dimensional construct or three unrelated constructs. X^2 is a numerical indicator of how paths between constructs fail to reproduce the responses predicted by a model [18]. The number of degrees of freedom (“df”) for each model reflects the number of items that are used by the model to predict outcome variables [18]. For each value that is predicted by a model, the number of degrees of freedom is reduced by one, but the amount of X^2 (misfit) will also be reduced by a level commensurate with the accuracy of the model. If a model is improved by the addition of paths between pairs of dimensions (i.e., the correlation of factors), then the ratio of X^2 to the change of degrees of freedom in the model would improve [18]. Based on X^2 and CFI difference, the correlated model provided a better fit ($p < 0.01$) for the data than did the uncorrelated and single dimension models. Thus, the most “true” model of the IT Identity Threat may be as a second-order construct [79]. These results are summarized in Table 7.

It should be noted that CFI scores above 0.9 are indicative of good fit [9] and RMSEA scores under 0.1 indicate good fit when parsimony is a consideration [12]. Based on these heuristics, Model 3, with its correlated factors, demonstrates less-than-ideal fit, but good fit when parsimony is considered. In light of the Model 3’s low RMSEA scores of 0.046 and 0.042 and its improvement of fit over Model 2, this study supports the conceptualization of the IT Identity Threat as a second-order construct.

Discussion

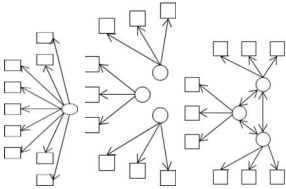
To address the research questions “*What is an IT Identity Threat?*” and “*How can it be measured?*” we conceptualized and operationalized a second-order construct and measure that integrates the social and task-related factors associated with resistance to IT. We believe that the IT Identity Threat is important due to 1) its ability to integrate a variety of manifestations of IT on social and work processes into a parsimonious model of resistance, and 2) its relevance in a world that is moving toward increasingly personal IT, making it important to understand resistance at the individual level.

Contributions to Theory

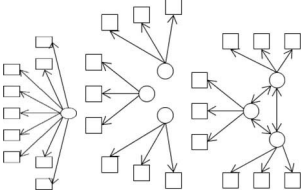
The definition and measure of the IT Identity Threat provide important theoretical contributions to the field of IS, and also to our reference disciplines. As a theoretical construct grounded in identity theory, with a rigorously tested measure, the construct offers a broader model of resistance to IT that builds on a tradition of prior qualitative work. The IT Identity Threat offers a variance-based approach to resistance to IT that integrates social, task-related, and personal perceptions of threat, caused by an IT.

Table 7. Dimensionality

Study 1 Model	χ^2	df*	CFI	$\Delta\chi^2$	RMSEA	ΔCFI
Model 1: One single dimension.	8189.07	53	0.79		0.270	
Model 2: Three uncorrelated factors.	1541.184	53	0.96	6647	0.115	0.17
Model 3: Three correlated factors.	277.85	50	0.99	1263	0.046	0.03



Study 2 Model	χ^2	df	CFI	$\Delta\chi^2$	RMSEA	ΔCFI
Model 1: One single dimension.	1916.22	54	0.18		0.413	
Model 2: Three uncorrelated factors.	118.63	54	0.97	1797	0.077	0.79
Model 3: Three correlated factors.	68.76	51	0.99	49	0.042	0.02



*For model fit purposes [18], the error terms between two items (S6, S7) were allowed to correlate, costing one degree of freedom.

The IT Identity Threat deals with the most fundamental element of resistance: the individual. Yet, it also accounts for factors normally treated on the group level. It is possible to conceptualize the intergroup conflict associated with threats to social identities according to its anticipated harm on a purely individual level, that is, according to individual perceptions of self-worth that are based on group membership [84]. In this regard, it answers the call for research that reconciles individual-level resistance with group-level phenomena [47].

For the field of IS research, this new construct provides a parsimonious causal factor that helps explain resistance to IT. Existing process models offer rich insight into how resistance and other behaviors evolve [59], but such insight may be expanded by engaging in variance-based research [50]. Variance-based models afford the opportunity for researchers to explore and collect data for a range of variables and theoretical relationships [66]. Thus, they afford progress based on the elaboration of testable models over time [87]. Moreover, because variance-based research may explore a phenomenon across a variety of contexts, it may serve as the basis of inferences about the general population of IT users [10,49]. Some have tried to incorporate numerous varied perceptions that may cause resistance to IT into a variance-based model. However, to do so without careful integration of theory raises the risk of “ballooning” explanations and contradictory approaches. For example, the work of Kim and Kankanhalli [44] used a variance-based approach, and also explored social factors along with purely individual perceptions of the IT artifact. However, their work only presents switching costs as the mediator of a collection of perceptual measures that already have well-studied relationships with either use or resistance to IT. To provide novel explanatory power requires the incorporation of novel theoretical logic. In addition, offering a collection of causal variables with limited theoretical justification for inclusion in or exclusion from a model may result in models that lack parsimony, as future researchers incorporate more and more variables [6]. A better approach is one that is parsimonious, with the simplicity and predictive power of a variance-based model along with carefully integrated theory from prior process models. The IT Identity Threat recognizes and integrates a broad range of perceptions into a single construct through their effects on the individual. Thus, it may serve as a foundation for building a cumulative tradition in this area.

In addition to integrating social factors with perceptions of the IT artifact, the IT Identity Threat addresses the emergence of more personal forms of IT. ITs are evolving and becoming more personal in their use [75], and technologies are now capable of changing the self in addition to impacting work tasks and social structure [20]. Thus, future resistance to IT is more likely to be the result of personal factors, and our field’s understanding of such factors is only just emerging [20]. As a result, in addition to explaining the effects of perceptions of the artifact in concert with social perceptions, the IT Identity Threat benefits the field by integrating these perceptions with perceptions of the self alone.

Finally, as a measurable identity threat construct, the IT Identity Threat may contribute to our field's reference disciplines. Previous works in sociology and psychology have proposed the existence of identity threats [64, 67, 68, 94]. However, they have offered broad and often conflicting definitions of identity threats and have not presented any means of measuring identity threats [68]. This work offers a reliable and valid measure of the IT Identity Threat—the first operational measure of an identity threat of any type—which expands and complements the qualitative tradition of identity research. It also informs our reference disciplines by demonstrating that the three types of identity (social, role, and person), as well as the three types of self-esteem (worth, competence, and self-authenticity), are conceptually related and interdependent. The correlations among our dimensions indicate that while the three types of identity (social, role and personal) have theoretically independent sources, they may be related. This finding supports the theoretical proposition that an identity of one type (e.g., a work role, such as “professor”) is frequently tied to one or more identities of another type (e.g., a social membership, such as “faculty member”) [60].

Implications for Practice

The IT Identity Threat provides novel insight into resistance to IT; in doing so, it affords new approaches to treating resistance for management. Existing literature advises increasing usability [51] and demonstrating managerial support for IT implementations [32] to encourage system use instead of resistance. In other words, academics prescribe that management make an IT less threatening or more appealing to encourage widespread adoption. There are times, however, when usability improvements may be insufficient to prevent a groundswell of resistance. For example, doctors in the UK's National Health Service have resisted mandated ITs, based on perceptions of those ITs as being part of “big IT” and instruments of bureaucratic control; efforts by management to encourage use may have been counterproductive as a result [37].

By contrast, the IT Identity Threat approach prescribes expanding users' identity, rather than changing perceptions of the IT, to avoid resistance and even encourage use. Specifically, when individuals are presented with an identity threat, they may seek an alternative identity in order to return to a cycle of identity confirmation and self-esteem growth [68]. For example, when faced with a task that threatens self-beliefs of competence, individuals may be motivated to learn new skills to expand the self rather than oppose the task. For example, when an organization shifts work tasks from a paper-based to an online system, some individuals may initially experience stress related to how well they use the online system, and how long it has been since they had held a “learner” identity. Nevertheless, they may be motivated to become a learner of the online system, if they ultimately believe that they can become a “master” of the online system. Doing this, however, requires personal change.

Management should understand that the implementation of an IT may result in the subjective loss of status, competence, and self-authenticity in the workplace, reflected by diminished self-esteem among workers. However, they are also advised that steps may be taken to help workers adapt and build new identities in the face of a threatening IT. Interventions designed to develop self-esteem among individuals by building identity have focused on three complementary tactics: reframing, recalculating, and refocusing [56]. Reframing involves infusing a workplace identity with positive meaning by communicating the necessity and dignity of an otherwise less-than-ideal work role. Recalculating involves changing how individuals measure the value of their work. Rewarding learning goals rather than production goals may be helpful in this regard. Finally, refocusing shifts attention toward the positive characteristics of a new or changed identity. This can take the form of emphasizing a shared identity that allows individuals to provide legitimacy and social support to each other as they become learners of a new IT together [4]. These tactics may help individuals like the professor in our vignette overcome the challenge of using an information system by emphasizing the value that learning such a system provides for their colleagues and organization. In the right light, trying to learn to use a threatening system may be an expression of *esprit de corps*, as members of an organization suffer together to master that system, verifying social identity. It could also be a demonstration of ability and competence, verifying role identity. Working hard to learn and help others master the system's features may invoke virtues such as perseverance and generosity, verifying person identity.

Directions for Future Research

This work investigates how ITs threaten individuals by changing their work lives and thus changing themselves. Our field's understanding of resistance to IT would benefit from considering individuals who do not want to be changed by ITs. By introducing a definition and measure of the IT Identity Threat, we hope to take the first step in broadening our field's approach in this regard.

Future researchers may leverage the theoretical richness and practicality of the identity threat framework [68] to pursue valuable lines of inquiry. Avenues of future research related to the IT Identity Threat include exploring moderators for the causes of resistance to IT, based on the identity literature. The identity threats are thought to be mitigated by individual beliefs about the self, such as the perceived ability to adapt to the threat source, and social beliefs, such as whether other individuals share perceptions of threat [68]. Thus, while an identity threat may cause resistance, it may also cause individuals to undergo change [95]. Individuals who believe they are capable of mastering a threatening IT, and thus mitigating its associated threat are more likely to eschew resistance in favor of becoming a proficient user of that IT. For example, Pratt, et al. [74], found that medical professionals are capable of incorporating unappealing work tasks into their professional identities, even though such tasks were perceived as "violating" those same

identities. Similarly, Coppola et al. [28], observed that many university professors who initially felt threatened by an online teaching system coped by adopting new “digital Socrates” personas. Thus, the IT Identity Threat may serve as an impetus for the personal change and learning that is associated with high levels of computer self-efficacy [26]. On the other hand, if individuals believe that others will collectively resist the source of an identity threat, they may be more likely to engage in resistance themselves. This is analogous to Lapointe and Rivard’s [47] observation that individuals are likely to resist an IT when they perceive group-level resistance.

Thus, using insights from identity threat research in concert with our own literature, future researchers may explore ways to stem resistance and possibly even leverage the IT Identity Threat as a motivating factor that may cause individuals to learn to use a threatening IT.

In addition, it is important to note that the studies presented in this work are limited to two contexts within the broader context of higher education. The field of IS could benefit from studies based on the IT Identity Threat in different contexts.

Limitations

We employed surveys to collect data across our studies⁴; thus, the possibility that our method of data collection was a factor in our results must be addressed. Accordingly, we examined our data for evidence of common method bias by evaluating correlations between unrelated marker items (found in Appendix E) and the measures of our constructs [24, 71]. These correlations ranged from 0.023 to 0.001, indicating that common method bias was not a factor in our results.

We note that the correlation between loss of competence-based self-esteem and loss of authenticity-based self-esteem in our first study is very high (0.779, see Table 6). Even in a formative second-order model, however, this may be found when a level of one dimension can influence the levels of one or more other dimensions [73]. Person identity is thought to influence the selection of workplace roles, and role-related experiences are impactful on person identity over time [17]. Thus, phenomena that impact person identity are likely to impact role identity, and vice-versa [17]. Nevertheless, a comparison with the average variance explained of each construct (in Table 6) indicates that they are distinct: related, but not the same thing. Additionally, since our construct’s formative dimensions are correlated, we estimated variance inflation factor scores of each dimension. The largest of these was 2.8 (see Appendix F for all scores), which is below the threshold of concern recommended by Petter et al. [70] of 3.3 for formative indicators.

Finally, we present the IT Identity Threat as a very broad construct that may explain a broad range of resistance behaviors as caused by loss of self-esteem. In doing so, we infer that it may subsume resistance factors that affect individuals’ self-esteem. However, we do not claim that it subsumes all sources of resistance to IT. For example, temporary emotional states and inhibitors such as computer anxiety may discourage IT use in general [90, 92]. When a system contributes to

users' problems rather than solve them, individuals may resist that system without experiencing identity threat. The IT Identity Threat may contribute to research on factors such as these, but is not intended to replace them.

Conclusion

As technology continues to infuse individuals' lives, our research must explore the ways it is changing lives in often harmful ways. Toward that end, this paper develops, defines, and measures a new construct—the IT Identity Threat—which captures the potential negative effect that IT can have on identity. Future research based on the IT Identity Threat will be more valuable as ITs influence what it means to be human. The IT Identity Threat can help explain why future generations of ever-more intrusive and personal technologies may be rejected by some users. However, because identity is malleable and can be re-formed, it also may point the way toward helping individuals adapt in order to realize the greatest possible benefit from emerging technologies.

Supplemental File

Supplemental data for this article can be accessed on the [publisher's website](#)

NOTES

1. The verification of identity is an event described in the identity literature. It can be defined as an individual's observation, either direct or through perceptions communicated by others, which confirms the self-beliefs that comprise an identity. In other words, when an individual perceives that they are the kind of person that they believe they are, their identity is verified [17] pp. 50-60.

2. This university was recently ranked in the top 20 Public Universities in the well-known US News and World Rankings of Best Colleges.

3. It must be noted that while the list service sent invitations by email to all students and faculty, it is not known how many of those invitations were treated as spam and ignored. According to the list service administrator, our response rate was typical.

4. Our sample frames were selected with resistance in mind. We surveyed populations of users that had vocally expressed discontent with the technology. For example, ePortfolio had been subject to criticism in the student newspaper and critically evaluated in conference papers. Also, post-implementation, iRoar had been the subject of committee meetings held by the CIO to address faculty concerns about the system. Consequently, while we expected variance in our sample frame, we also anticipated high levels of threat perceptions were likely among subjects who selected to complete our survey.

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